

Assignment 1

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D20A

Explore following 4 websites and answer the questions

- bitcoin.org
- Blockchain.com
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom?
Satoshi Nakamoto
2. Bitcoin transactions are recorded on which data structure?
Blockchain
3. Bitcoin uses which consensus mechanism?
Proof of Work
4. The smallest unit of Bitcoin is called?
Satoshi (sat)
1 Bitcoin = 100,000,000 Satoshis
5. Bitcoin is a **peer-to-peer** type of what system?
Electronic cash

Wallets

6. A wallet stores the user's Private key.

7. The recovery phrase used in wallets is called a Seed phrase.

8. Hardware wallets store keys offline (online/offline).

9. Bitcoin addresses are derived from Public keys.

10. What is required to sign a transaction? (public/private)
Private

Security

11. Bitcoin prevents double-spending using blockchain.

12. Bitcoin is secured using asymmetric cryptography (asymmetric/symmetric).

13. To run a Bitcoin node, the software required is Bitcoin core.

14. The process of verifying transactions is called validation.

15. The Bitcoin network is described as decentralized (centralized/decentralized).

Mining

16. Miners solve cryptographic puzzles to add blocks.

17. Mining rewards a block with newly created bitcoins.

18. The block time target for Bitcoin is 10 minutes.

19. Mining difficulty adjusts every **2016** block.

20. Miners compete to find a valid block hash (hash/signature).

Blockchain Structure

21. Each block contains a Merkle root.

22. Blocks are linked using the previous block's hash.

23. A Bitcoin transaction contains inputs and outputs.

24. Unspent transaction outputs are called UTXOs (Unspent Transaction Outputs).

25. The first-ever Bitcoin block is called the Genesis block.

Network

26. A full node stores the entire blockchain Ledger.

27. Bitcoin uses which protocol for communication? (TCP/P2P)
TCP

28. A light wallet uses SPV mode (SPV/full).

29. Nodes relay information across a Peer-to-peer network.

30. The maximum Bitcoin supply is 21 million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height: **960833**
- Block hash:
000000000000000000031f27be35daba44eecd10b82776d82f77b8e80e976c7
- Timestamp: **2026-08-03 12:01:30 GMT +5.5**
- Number of transactions: **5516**
- Miner/pool name: **unknown**

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction.
f4c76d17dc68e482e84bb18248d705fc3d4e6ab55dfecad68a309610edf26c17
11. Number of inputs: **1**
12. Number of outputs: **1**
13. Transaction fee: **0.00000110 BTC (1.01 sat/vB)**

14. How many confirmations does it have?
960,835

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.
bc1qylrcfj22ff8hv8szj4d3j0aqyzrwm9gl0x8z35

16. Record:

- Total BTC received: **51.86941224 BTC**
- Total BTC sent: **51.86941224 BTC**
- Current balance: **\$0.00**
What is the role of a full node?

List any two developer resources available on bitcoin.org.

Developer Guides: Covers topics such as Blockchain, Transactions, Wallets, Mining, P2P Network, Payment Processing, and Contracts.

Reference Documentation: Provides technical details including the RPC API Reference, Blockchain Reference, Transaction Reference, Wallet Reference, and P2P Network Reference.

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC: **\$62,716.48**
- 24-hour % change: **1.12%**
- Market capitalization: **\$1.25T**
- Circulating supply: **20.06M BTC**
- Total/maximum supply: **20.06M BTC**
- **Task C2: Historical Data**
- What was the price of Bitcoin one year ago? **\$113,730**

- Note the highest and lowest prices in the last 30 days.
Highest Price: ~\$66,700 USD
Date: Late July 2026 (specifically around July 23–25, 2026)

Lowest Price: ~\$58,000 USD

Date: Early July 2026 (recorded around July 3–5, 2026 during the early July)

Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile?

Bitcoin is considered **volatile** because its price can rise or fall significantly in a short period due to factors such as market demand and supply, investor sentiment, news, regulations, and macroeconomic events.

List two investment risks mentioned.

High price volatility leading to potential financial losses.

Regulatory uncertainty, where changes in government policies or laws can affect Bitcoin's value and usage.

♦ Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Bitcoin.org gave the clearest understanding because it provides well-structured explanations, beginner-friendly guides, and official documentation on Bitcoin, wallets, mining, transactions, and blockchain concepts with simple language and practical examples.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without a central authority by storing transactions in blocks that are cryptographically linked using hashes. Every transaction is verified by a decentralized network of nodes through a consensus mechanism such as Proof of Work. Once a block is added to the blockchain, altering its data would change its hash and invalidate all subsequent blocks, making tampering extremely difficult. Since every full node maintains a copy of the blockchain and independently verifies transactions, no single organization or individual controls the network, ensuring transparency, security, and trust among participants.